

ARBITRATOR

Offline Online Judge System for University Programming Labs

Problem Statement

- ✕ Programming labs need a simple way to run coding contests
- ✕ Internet is not allowed in contests



Solution: An offline judging system that runs entirely within LAN, providing automated code evaluation and real-time feedback.

Project Overview

Arbitrator is an offline, LAN-based online judge system designed for competitive programming. It will work as a permanent solution to existing problems in Intra-Contests or Lab Contests.



Offline judging system

No internet connection required



Built in IDE

Code and test instantly



Live Standings

Real-time contest rankings



Code Visualizer (out of scope)

Understand execution visually

The system runs on the instructor's computer as a server, and students connect using a desktop client from lab computers.

What the Application Will Do

- **Arbitrator** allows instructors to create programming contests and students to submit solutions from lab computers.
- **The system** automatically compiles, runs, and judges submitted code against test cases.
- **Students** receive instant verdicts and can track their progress on a live leaderboard.

Verdict Types

AC — Accepted

WA — Wrong Answer

TLE — Time Limit

CE — Compile Error

RE — Runtime Error

MLE — Memory Limit

Plus live leaderboard and contest timer

Key Capabilities:

- Automatic compilation, execution, and judging of submitted code
- Built in IDE for Students
- Live **leaderboard** with contest timer
- Support for C, C++, Java, and Python

Main Users of the System



Student

- Login to the system
- View contest problems
- Write using Arbitrator IDE
- Submit Code
- Receive verdicts
- View submission history
- Check leaderboard



Instructor / Admin

- Create and manage contests
- Upload problem set
- Monitor submissions
- Send announcements
- View/Freeze Leaderboard



Judge System

- Compile submitted code
- Run code with test cases
- Use checker
- Deliver verdict
- Update leaderboard

Planned Features

- ❖ Registration and login
- ❖ Role-based access control
- ❖ **Contest** creation and timer
- ❖ Problem upload
- ❖ Integrated **code editor**
- ❖ Code submission (C, C++, Java, Python)
- ❖ Verdict generation (AC, WA, TLE, MLE, CE, RE)
- ❖ **Real-time** verdict delivery
- ❖ Live **leaderboard**
- ❖ Live announcements
- ❖ Submission **history**

Student Interface Features

- ➡ Login screen
- ☰ Problem list
- 📄 Problem statement viewer
- ⌕ Code editor with highlighting
- 🗑 Language selection
- Submit button
- 🕒 Live contest timer
- ✅ Verdict notification
- 🕒 Submission history
- 🏆 Leaderboard view
- 📢 Announcement pop-ups

Student Workflow

Login → Select Problem → Write Code → Submit → Get Verdict → View Leaderboard

Admin Panel Features

- Create, edit, and manage contests
- Start, freeze, and end contests
- Upload problem packages
- Manage registered users

- View all submissions
- Monitor judge queue
- Rejudge submissions if needed
- Broadcast announcements
- View final standings and reports

Technologies and Tools

Java	Main programming language
JavaFX	Desktop client interface
Spring Boot	Backend server application
PostgreSQL	Database storage
Spring Security + JWT	Authentication and access

WebSocket / STOMP	Real-time updates
Java ProcessBuilder	Code sandbox execution
g++, JDK, Python	Compilers / Interpreters
Maven / Gradle	Build tool
RichTextFX	Syntax highlighting

Expected Benefits



No Internet Needed

Works entirely offline within the university LAN



Easy Management

Makes lab contests easier to organize and run



Saves Time

Reduces instructor workload with automation



Quick Feedback

Students get instant verdicts on submissions



Multi-Language

Supports C++, Java, and Python



Fair Scoring

Automated judging ensures consistent evaluation



Data Security

All data stays inside the university LAN

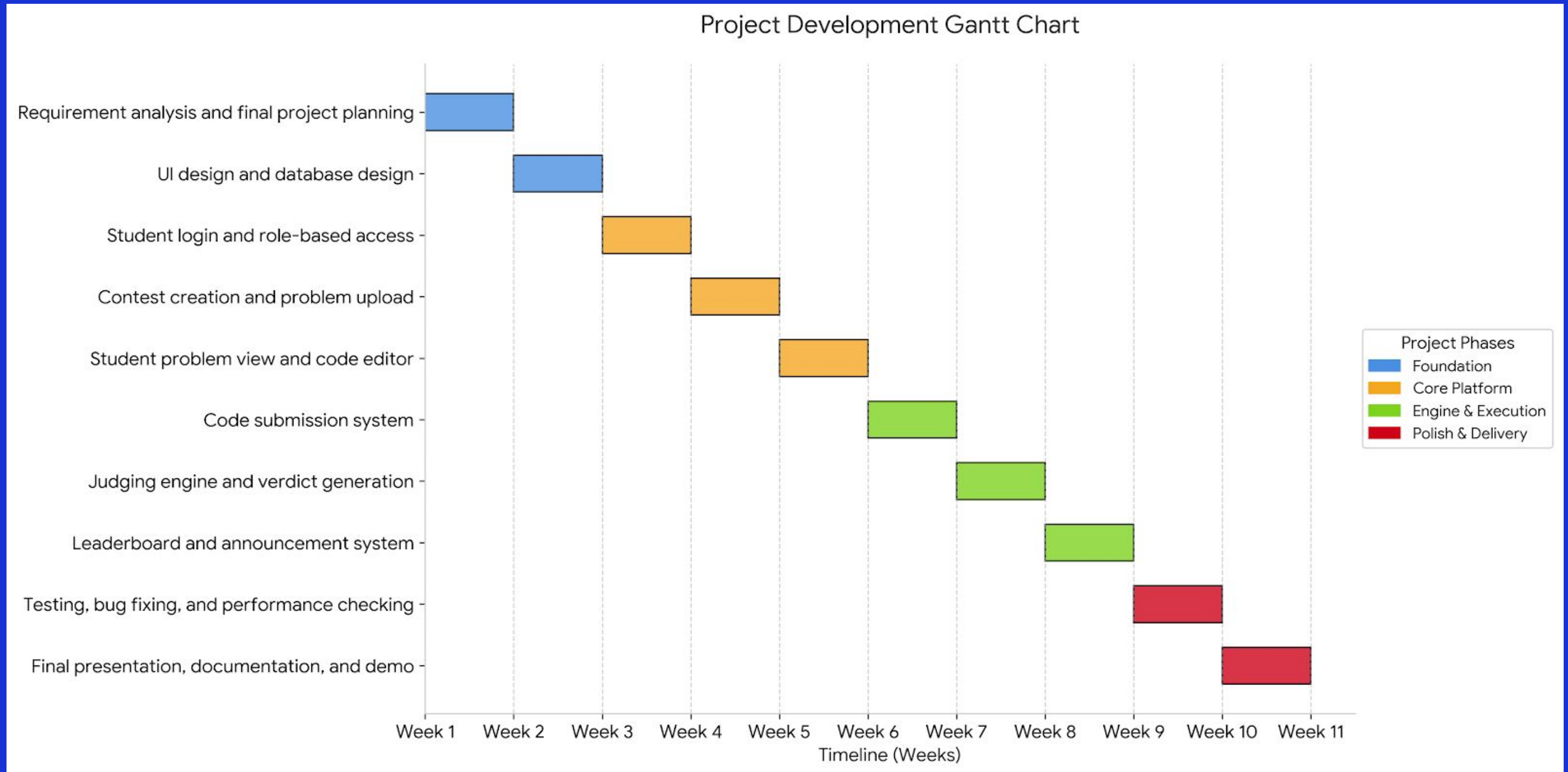


Versatile Use

For practice, exams, and competitions

Arbitrator brings the power of online judging into a secure, offline environment tailored for university programming labs.

Tentative Project Timeline



Expected Final Outcome



A working offline online judge system for university programming labs where students can join contests, submit code, and receive automatic verdicts — all without internet access.



For Students: Join contests, submit code, and receive instant automatic verdicts



Deployment: Runs entirely on the university LAN with no external dependencies



For Instructors: Manage contests, problems, submissions, and reports from a local server



Languages: Full support for C++, Java, and Python programming languages

Thank You

Arbitrator makes programming lab contests easier, faster, and more organized.